

Name:	ID:
-------	-----



Brac University

Semester: Fall 2023
Course No: CSE251

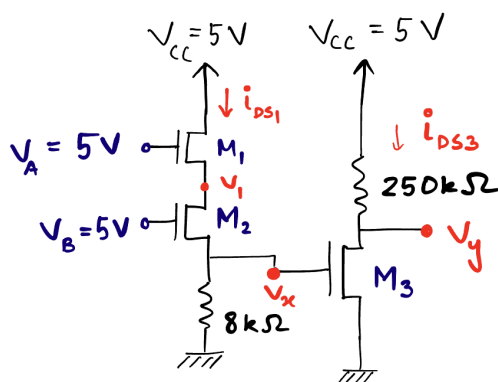
Set - A

Final Examination
Full Marks: 30
Time: 1.5 hours
Date: Dec 15, 2023

Course Title: Electronic Devices and Circuits
Sections: 1 to 24

Answer any 3 questions. All the questions carry equal marks.

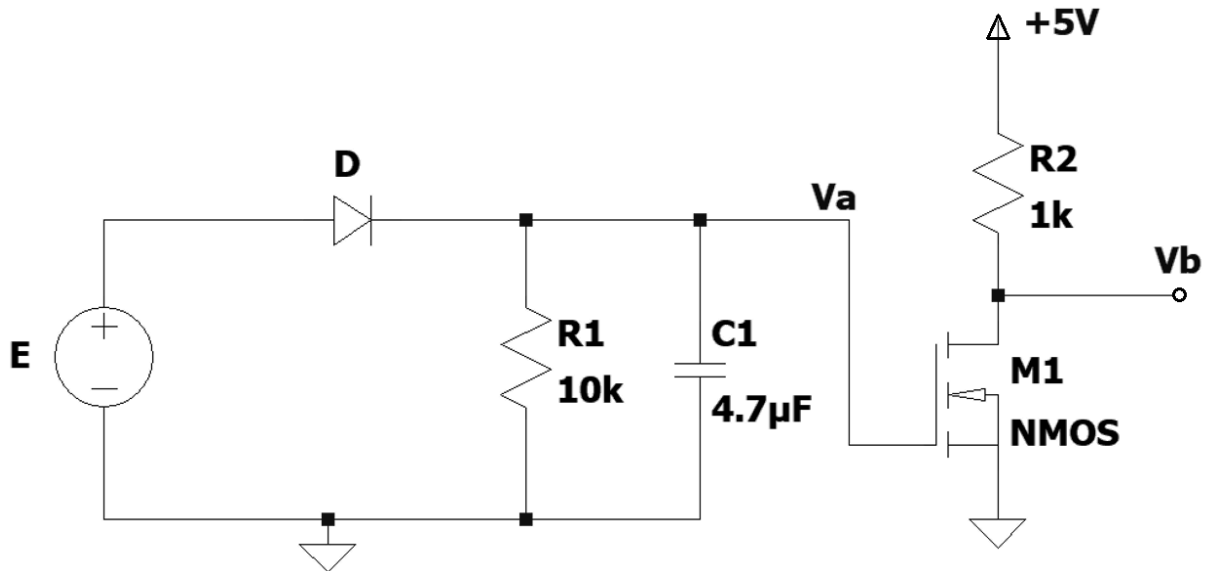
Question 1



For the following circuit with MOSFETs, assume that $V_T = 1\text{ V}$, $k_n' = 50\text{ }\mu\text{A/V}^2$ and the aspect ratio **W/L** is 1 for all the MOSFETS.

- If the gate voltage to **M3** MOSFET, $V_x = 0.938\text{ V}$, and MOSFET **M2** is in triode mode, find the voltage V_1 . [3]
- Determine the value of V_y . [2]
- If **W/L=2**, then $V_1 = 2.1454\text{ V}$. Find V_x for this case. [2]
- Will there be any changes in the output voltage at V_y for **W/L = 2** from that in '**part-b**'? Explain briefly. Also, comment on whether a larger or smaller value of **W/L** ratio is preferred in this case. [2+1]

Question 2

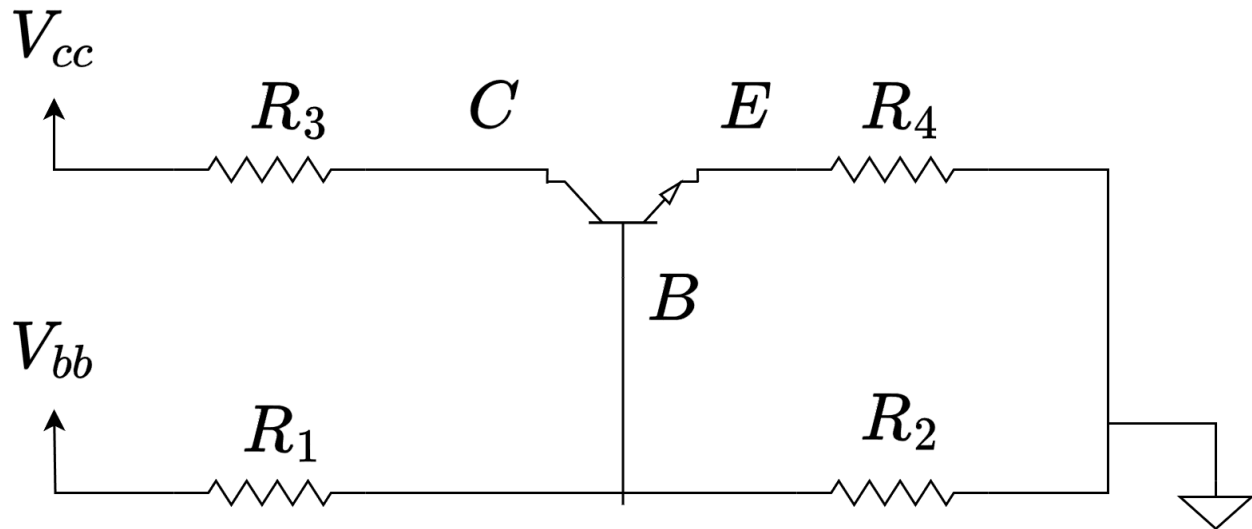


The Half-wave rectifier circuit has been cascaded with an inverter. You are given the following information about the circuit: **[For HW rectifier: $V_o = V_{dc} \pm V_{ripple} / 2$]**

$E = 5 \sin(400\pi t) V$	$K_n = 2 mA/V^2$	$V_T = 1 V$	$V_{Do} = 0.7 V$
--------------------------	------------------	-------------	------------------

- Determine** the peak-to-peak ripple value of the half-wave rectifier output, V_a . What is the lowest value of V_a ? [3+1]
- At the lowest value of V_a , **determine** the output V_b of the inverter. [4]
- Now disconnect the capacitor. In this case, **find out** the voltage V_b for the lowest value of V_a . [2]
- [BONUS]** Draw an approximate plot of V_b without the capacitor connection. [2]

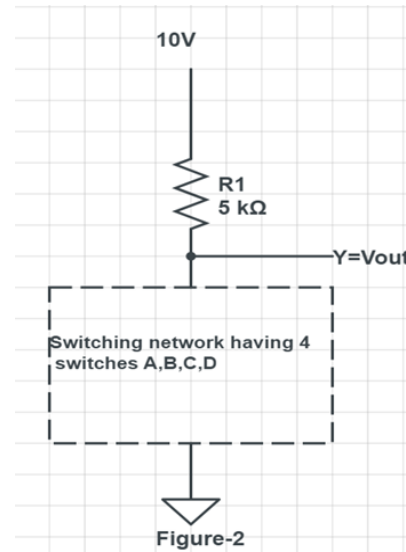
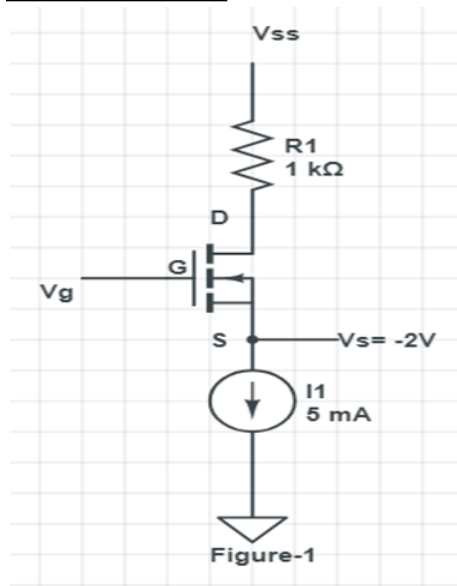
Question 3



In the circuit above, $V_{bb} = 5\text{ V}$, $V_{cc} = 15\text{ V}$, $R_1 = 20\text{ k}\Omega$, $R_2 = 80\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$ and $R_4 = 1\text{ k}\Omega$, $\beta = 100$.

- Draw** the equivalent circuit of BJT during saturation and active modes. [1]
- Analyze** the following circuit to **calculate** the values of I_b , I_c , I_e , V_{CE} and V_C using the method of assumed states. You must validate your assumptions. [6]
(Hint: find the Thevenin equivalent of the bottom circuit from the B terminal and ground)
- If V_{bb} is changed **from 5V to 5.1V**, calculate I_b , I_c , I_e , V_{CE} and V_C again. Now for a 0.1V increase in input V_{bb} , what is the change of I_c ? [3]
Use the equation, $\Delta I_c = I_{C(NEW)} - I_{C(OLD)}$
- [BONUS]** Explain any use case of the differences in voltage increase between input and output. What could the use case be to such a phenomenon? [1]

Question 4



- a) For the MOSFET in Figure-1, $V_T = 1\text{ V}$ and $k_n' = 2\text{ mA/V}^2$. Assume that the MOSFET is in **Saturation mode**. Determine the gate voltage, V_g and the minimum supply voltage, V_{ss} to operate the device in this condition. [Hint: $V_{OV} = V_{DS}$] [3+2]
- b) In Figure-2, the MOSFET switching network turns ON the following conditions MUST be fulfilled -

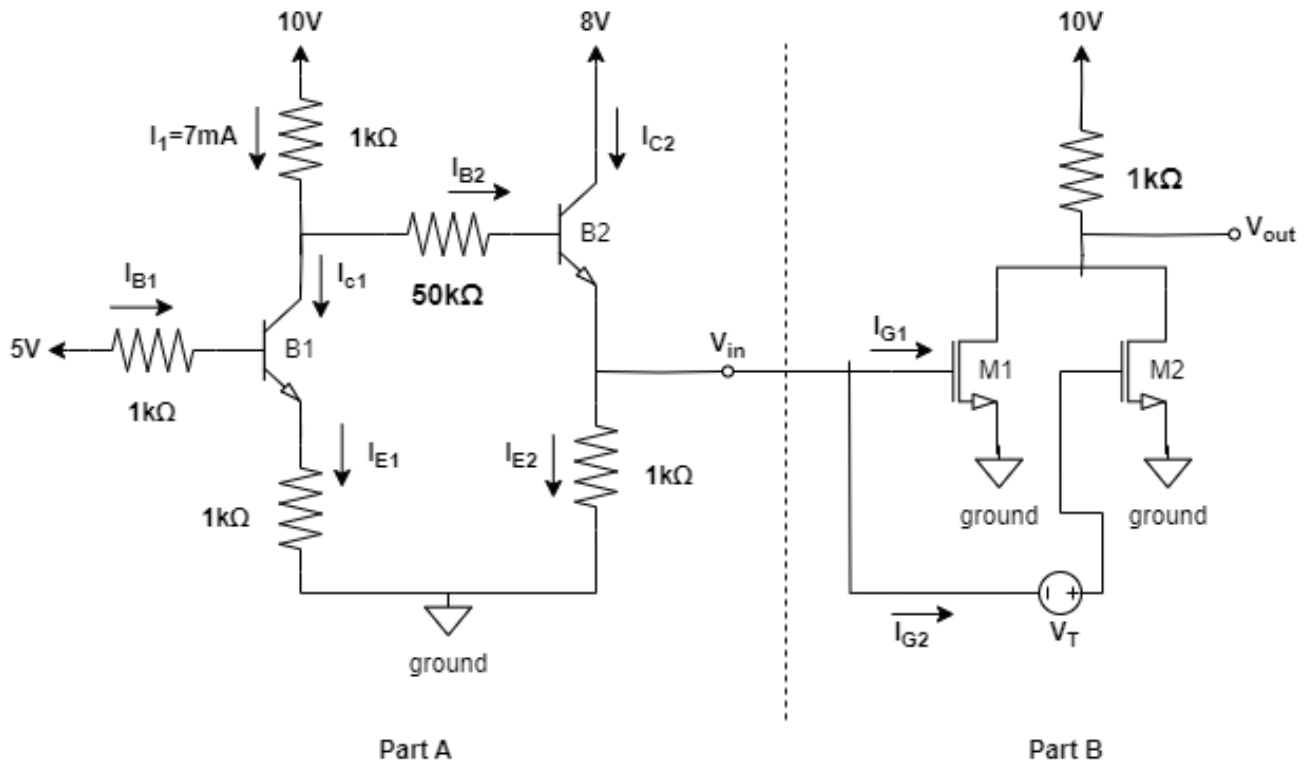
- 'A' MUST be ON
- 'B' and 'C' MUST be ON or 'D' MUST be ON

Deduce the logic function, F, using Boolean variables A, B, C, and D to implement these conditions. [2]

- c) Assume, only A and D switches are ON in the switching network of Figure-2. **Find out** the output voltage, V_{out} . Let, $R_{ON} = 0.1\text{ k}\Omega$ and use the SR model. [3]

- d) **[BONUS]** In Figure-2, assume the input on a NOT gate is connected to V_{out} . **Explain** mathematically whether it will be possible to turn OFF the NOT gate when V_{out} is LOW. Assume, $V_T = 0.5\text{ V}$, $V_{out} = V_{out(LOW)}$ of 'part-b'. [2]

Question 5



In the circuit above, the BJTs have the following specification: $\beta=100$, Forward Active Region: $V_{BE} = 0.7\text{ V}$, $I_C = \beta I_B$, Saturation Region: $V_{BE} = 0.8\text{ V}$, $V_{CE} = 0.2\text{ V}$, or the MOSFETs: $V_T = \text{Threshold Voltage of M1 and M2}$.

- Determine i_{g1} and i_{g2}
- Justify** why the SR model of MOSFET is more efficient than the S model ? [1]
- Assume, B1 and B2 are in the **Saturation region**. Calculate i_{c2} .
- Assume, B1 is in the **Forward Active region**. Calculate V_{in} .
- Draw the VTC** of Part- B assuming, $V_T = 8\text{ V}$. [Use S model of MOSFETs]

Equations for MOSFET

$$I_D = 0, \text{ if } V_{GS} < V_T$$

$$I_D = k \left[(V_{GS} - V_T) V_{DS} - \frac{1}{2} V_{DS}^2 \right], \text{ if } V_{GS} \geq V_T \text{ and } V_{DS} < (V_{GS} - V_T)$$

$$I_D = \frac{1}{2} k (V_{GS} - V_T)^2, \text{ if } V_{GS} \geq V_T \text{ and } V_{DS} \geq (V_{GS} - V_T)$$